

# SIMATS ENGINEERING

## CO4-ASSESSMENT TOOL2- ARTICLE REVIEW

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1709

### COURSE OUTCOME COVERED

**CO4:** Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 10%

**Time Duration: 1 Hour**

**QUESTIONS: 4**

**Total Marks:40**

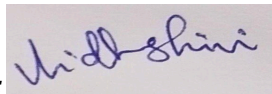
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### Code of Conduct:

*I Vidhushini.T.S (192511061) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:**



### **INSTRUCTIONS**

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

**Submit:** (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

### **Task 1: Article Summary**

**(a)** Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

**(b)** Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

### **Task 2: Methodology Analysis**

**(a)** Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

**(b)** Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

### Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

### Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

### RUBRICS FOR CALCULATION:

| Criteria                 | Weightage (%) | Level 4 (Exemplary)                                                                             | Level 3 (Proficient)                                 | Level 2 (Developing)                     | Level 1 (Beginning)                      |
|--------------------------|---------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------|------------------------------------------|
| Understanding of Article | 20%           | Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments | Shows good understanding with minor gaps             | Partial understanding; misses key ideas  | Lacks understanding of the article       |
| Summary                  | 20%           | Provides concise and accurate summary highlighting all key points                               | Covers most key points with minor omissions          | Incomplete or partially accurate summary | Poor or incorrect summary                |
| Critical Analysis        | 25%           | Provides insightful critique with strong reasoning and evaluation                               | Provides reasonable analysis with some justification | Superficial analysis; limited evaluation | No meaningful analysis                   |
| Use of Evidence          | 15%           | Effectively uses examples, data, or references to support review                                | Uses some supporting evidence with minor gaps        | Limited or weak supporting evidence      | No supporting evidence provided          |
| Organization             | 20%           | Well-structured, clear, and coherent writing with proper language and flow                      | Generally clear with minor issues                    | Poor organization; lacks clarity         | Disorganized and difficult to understand |

## ARTICLE REVIEW

Selected Article :

Title: Applying reinforcement learning and tree search to the unit commitment problem

Authors: Patrick de Mars and Aidan O'Sullivan

Journal: Applied Energy

Publisher: Elsevier

Volume: 302

Year: 2021

Article Number: 117519

DOI: 10.1016/j.apenergy.2021.117519

Topic: Reinforcement Learning

Application Area: Power system operation and electricity generation scheduling

The article applies reinforcement learning to the unit commitment problem, where power generators have to be scheduled to meet electricity demand while keeping operating costs low. The authors combine reinforcement learning with a tree-search method to make the solution more suitable for larger power systems.

### TASK 1 – ARTICLE SUMMARY

1(a) Article Citation, Domain and ML Problem

Full Citation

de Mars, P., & O'Sullivan, A. (2021). Applying reinforcement learning and tree search to the unit commitment problem. *Applied Energy*, 302, 117519. Elsevier. DOI: 10.1016/j.apenergy.2021.117519.

Domain/Application Area

The article belongs to the power systems and energy management domain.

It focuses on day-ahead unit commitment, which means deciding which electricity generators should be switched ON or OFF and scheduled for operation in advance.

Machine Learning Problem

The main ML problem is applying Reinforcement Learning (RL) to a large-scale unit commitment problem.

The difficulty is that as the number of generators increases, the number of possible actions grows very quickly. This makes directly applying normal RL methods difficult for larger power systems.

## 1(b) Key Objective and Research Gap

### Article Summary

The main objective of the article is to develop a reinforcement learning-based approach for solving the unit commitment problem in power systems. Unit commitment involves selecting the operating schedule of generators while satisfying electricity demand and trying to reduce operating cost. Traditional optimization methods such as mixed-integer linear programming can solve these problems, but their computational requirements can become high when the problem becomes large and uncertain. Reinforcement learning provides another approach because an agent can learn useful decisions by interacting with a simulated environment. However, the number of possible actions increases exponentially as the number of generators increases. The authors address this problem by combining a learned RL policy with tree search. The RL policy is used to guide the search and reduce the number of possible branches that need to be examined. The method was tested on simulated power systems containing between 5 and 30 generators using demand and wind data based on the GB power system. The results showed that guided tree search could handle larger systems more efficiently while maintaining competitive operating costs.

### Research Gap

The main gap identified by the authors was that previous RL approaches for unit commitment were generally limited to small power systems, with previous applications considering systems of up to about 12 generators. The large action space made direct application difficult for larger systems.

### How the Authors Addressed the Gap

The authors proposed guided tree search, where the RL policy guides the tree search and reduces its branching factor. This allows the method to consider larger systems without examining every possible action.

## TASK 2 – METHODOLOGY ANALYSIS

### 2(a) ML Technique and Dataset

#### Machine Learning Technique

The main technique used is Reinforcement Learning.

In reinforcement learning:

- An agent interacts with an environment.
- The agent observes the current state.
- It selects an action.
- It receives a reward.
- The agent gradually learns a policy for selecting better actions.

In this article, RL is used to learn useful generator commitment decisions.

The authors then combine the learned policy with tree search.

#### Guided Tree Search

The basic idea is:

RL Policy → Guide Tree Search → Reduce Branches → Find Good Solution

Instead of searching all possible combinations of generator decisions, the learned RL policy helps identify promising branches.

This is important because the action space becomes very large when the number of generators increases.

#### Dataset

The experiments use demand and wind data based on the GB power system.

The researchers simulated power systems containing:

- 5 generators
- 10 generators
- 15 generators
- 20 generators
- 25 generators
- 30 generators

The experiments were designed to test whether the proposed approach could generalize to larger systems and to unseen test profiles.

#### Data Preparation

The power-system data was used to create simulated unit commitment environments. The RL agent was trained through interaction with the simulated environment, while separate unseen profiles were used for testing.

This is useful because it checks whether the learned policy can work on situations that were not directly used during training.

## 2(b) Mapping to CO4

The assessment specifically asks for mapping the article to CO4 topics such as Inductive Learning, Decision Trees, Statistical Learning and Reinforcement Learning.

| CO4 Topic              | Relation to Article                                                   |
|------------------------|-----------------------------------------------------------------------|
| Inductive Learning     | The system learns useful decision patterns from previous interactions |
| Decision Trees         | Tree search is used, but it is not a Decision Tree classifier         |
| Statistical Learning   | Not the main technique                                                |
| Reinforcement Learning | Main technique used in the article                                    |

### Strength

One major strength is that the proposed approach is tested on systems with up to 30 generators, which is larger than many earlier RL-based unit commitment studies. The use of unseen test profiles also provides a useful test of generalization.

### Limitation

One limitation is that the method depends on a simulated power-system environment. Real-world power systems have many additional constraints and uncertainties. Therefore, good simulation results do not automatically guarantee the same performance in a live power grid.

## TASK 3 – RESULTS & CRITICAL EVALUATION

### 3(a) Key Results

The authors compared the proposed guided tree-search approach with an unguided tree search and with mixed-integer linear programming (MILP) approaches.

The main findings were:

1. Guided tree search reduced the computational complexity compared with unguided tree search.
2. The proposed method maintained similar operating-cost performance.
3. The method was tested on systems ranging from 5 to 30 generators.
4. The guided approach was able to solve unseen test problems.
5. The authors reported that guided tree search outperformed an MILP solution using reserve constraints.
6. The RL-based solutions produced operating strategies that were qualitatively different from MILP solutions.

### Comparative Table

| Method                        | Main Purpose              | Computational Performance                           | Cost Performance |
|-------------------------------|---------------------------|-----------------------------------------------------|------------------|
| Unguided Tree Search          | Search possible solutions | Higher search complexity                            | Good             |
| Guided Tree Search            | RL-guided search          | Better scalability                                  | Competitive      |
| MILP with Reserve Constraints | Traditional optimization  | Higher computational requirement for large problems | Competitive      |

The article's main contribution is therefore not simply obtaining a lower operating cost. It is making the search process more manageable for larger unit commitment problems.

### 3(b) Critical Evaluation

Are the Results Realistic?

The results appear reasonable because the authors tested different system sizes and also used unseen test profiles. Testing from 5 to 30 generators gives a better indication of scalability than testing only one small system.

However, the results should still be interpreted carefully because the experiments are based on simulated power-system conditions.

## Possible Biases

- The data is based on the GB power system.
- The experiments are performed in a simulated environment.
- Different countries may have different generator characteristics and operating constraints.
- The performance may change with different demand and renewable-energy patterns.

## Additional Experiments

1. Testing the method using data from different countries.
2. Testing more extreme renewable-energy conditions.
3. Comparing the method with more recent RL algorithms.
4. Testing larger numbers of generators.
5. Evaluating performance using real operational power-system data.
6. Studying the effect of unexpected failures or sudden changes in demand.

## 3(c) Real-World Applicability

The technique could potentially be used in:

- Electricity generation scheduling
- Smart grid management
- Renewable energy integration
- Power-system decision support
- Energy management systems
- Day-ahead electricity planning

The authors also suggest that the RL solutions have potential as a decision-support tool for human operators because the learned strategies can provide alternative operating solutions.

## Challenges in Industry

### 1. Data availability:

Reliable historical and real-time power-system data is required.

### 2. Computational requirements:

Training RL models and performing search can require significant computing resources.

### 3. Safety:

An incorrect generator schedule could affect power-system reliability.

### 4. Generalization:

A model trained on one power system may not perform equally well on another.

### 5. Human supervision:

For critical infrastructure, the system should assist human operators rather than blindly control the grid.

### 6. Changing conditions:

Demand, renewable generation and equipment conditions can change continuously.



## TASK 4 – REFLECTION & LEARNING OUTCOME

### 4(a) Three Key Insights

#### Insight 1 – Reinforcement Learning learns through interaction

I understood that reinforcement learning is different from normal supervised learning. Instead of being given the correct answer for every input, the agent learns by taking actions and receiving rewards.

Connection to CO4: Reinforcement Learning.

#### Insight 2 – Large action spaces can make RL difficult

The article helped me understand that reinforcement learning can become difficult when the number of possible actions becomes very large. In the unit commitment problem, adding more generators increases the number of possible decisions significantly.

The guided tree-search method provides a way to reduce this problem.

Connection to CO4: Reinforcement Learning and decision-making.

#### Insight 3 – Model performance is not only about accuracy

This article showed me that in real-world decision-making problems, performance cannot always be measured using only accuracy or F1-score. For a power-system problem, factors such as operating cost, computational complexity, scalability and reliability are also important.

Connection to CO4: Evaluation of machine learning models and decision-making systems.

### 4(b) Proposed Extension

#### Proposed Experiment

I would extend this research by testing the proposed RL and guided tree-search method using power-system data from multiple countries, including systems with different renewable-energy conditions.

For example, the experiment could compare:

GB Power System → Indian Power System → High-Renewable Test System

The same RL model could be trained and tested on these different environments.

#### Justification

This experiment would help determine whether the method is general enough to work outside the original GB-based simulation. It would also show how the method performs when demand and renewable-energy patterns are different.

The expected impact would be better understanding of the generalization capability of reinforcement learning for power-system scheduling. It could also make the approach more useful for practical smart-grid applications.

The article provides a useful application of reinforcement learning to a practical power-system optimization problem. The main contribution is the combination of an RL policy with tree search to overcome the large action space of the unit commitment problem. The experiments show that guided tree search can handle systems with up to 30 generators while maintaining competitive operating costs.

From a CO4 perspective, the article gives a clear example of how reinforcement learning can be applied to real-world decision-making. It also shows that developing an ML solution involves more than selecting an algorithm; scalability, computational cost, generalization and practical constraints must also be considered.